Pierce Lonergan

Software Engineer III · Data & Applied ML · Platform

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PROFESSIONAL SUMMARY

Most RAG systems fail at the pipeline, not the model. I have spent four years building the pipeline. Software Engineer III at JPMorganChase, building the streaming and batch data infrastructure Consumer & Community Banking runs on: reusable platforms across Kafka, Spark, Iceberg, Snowflake, and AWS. Alongside it I build retrieval systems in the open and payment infrastructure on Spring Modulith. Software Engineer to Software Engineer III in four years. B.S. Biochemistry with a Computer & Information Science minor, and a bias toward measurement and honest evaluation, including of my own results. Open to applied ML, data engineering, and platform roles.

TECHNICAL SKILLS

Streaming & Big Data: Apache Kafka, Apache Spark (Structured Streaming), Apache Iceberg, Apache Avro, Project Reactor, Cassandra, schema evolution, CDC (Debezium), exactly-once processing

Lakehouse & Warehouse: Snowflake, Iceberg lakehouse, Parquet, partitioning, data modeling, incremental sync

Applied ML & Retrieval: RAG, hybrid retrieval (BM25 + dense), ColBERT / late-interaction reranking, cross-encoder reranking, BGE embeddings, Qdrant, LLM application development, INT8 quantization

Cloud, Platform & Infra: AWS (S3, EMR, MSK, Glue, Kinesis, Lambda), Docker, CI/CD, hexagonal architecture, microservices, Spring Boot / Spring Modulith, Temporal, Keycloak, PostgreSQL

Data Governance: Canonical catalogs, semantic schema matching, entity resolution, lineage, data-quality validation

Languages: Python, Java, Scala, Groovy, Bash, SQL

PROFESSIONAL EXPERIENCE

JPMorganChase | Consumer & Community Banking, Data Engineering — Columbus, OH

Software Engineer III (Feb 2026 – Present) · Associate Software Engineer (Jan 2024 – Feb 2026) · Software Engineer (Apr 2022 – Jan 2024)

- Architected the reusable streaming-pipeline layer (declarative source/sink/transform factories, forward-compatible schema evolution, generic recursive flattening for nested data). Team throughput went from roughly 2 to 30 production pipelines per month over the two quarters after adoption; I built the framework, the number is the team's.
- Standardized pipeline lifecycle concerns once and reused them across every pipeline: checkpointing, error handling, and graceful shutdown.
- Shipped data-governance automation that replaced blank-page hand-mapping by stewards with machine-proposed mappings they confirm or correct, cutting weeks from approval cycles (product-owner estimate, not an instrumented baseline); now a production internal product. Drove schema-evolution standards, semantic schema matching, and entity resolution against a canonical catalog.
- Built high-performance Spark ETL with asynchronous processing and parallelization, and integrated Kafka for near-real-time distributed streaming, reducing latency and improving reliability and scalability.
- Modernized big-data systems with reactive paradigms (Project Reactor), pairing Kafka messaging with Cassandra reconciliation and Spark ETL, and migrated high-throughput systems to federated AWS.
- Led a hackathon team to 4th place globally; architects projected the platform would save roughly 80 engineer-hours per month.

SELECTED PROJECTS

NexusPay | Enterprise payment-operations platform — Personal project · Java

- 17-module Spring Modulith (Java 21, hexagonal) on HyperSwitch: a double-entry ledger holding its invariants under SERIALIZABLE, transactional outbox to Kafka via Debezium CDC so nothing is dual-written, fraud rules, FX, a dispute state machine, subscription billing, Keycloak SSO with maker-checker approvals, Temporal, Vault (PCI), Prometheus / Grafana SLOs. Built solo with an agent-assisted loop; the architecture, module boundaries, money invariants and test strategy are mine.

NexusMatcher | Semantic schema-matching system — Personal project · Python

- Multi-stage retrieval with late-interaction reranking and learned type projections (hybrid BM25 + dense, ColBERT, Qdrant). Reranks 100 candidates in 3.17 ms warm against 274 ms cold, a 94× speedup from precomputed token embeddings. P@1 is 1.0 on the in-domain benchmark and 0.29 on a harder 17-field set where P@5 holds at 0.76; 433 tests.

MOMENTUM-X | LLM-driven algorithmic trading system — Personal project · Python

- End-to-end intraday trading system on the Alpaca API (phase-based stop-loss, live daemon, full observability), plus mx-arena, a local exchange simulator for backtesting. Pre-registered the hypothesis and the kill criteria, validated out-of-sample with walk-forward evaluation, then ran an adversarial audit against my own conclusion: the multi-agent LLM ensemble consensus came out anti-predictive. I recorded the negative result and killed the feature I had spent months building.

EDUCATION

The Ohio State University | B.S. Biochemistry, minor in Computer & Information Science — Aug 2016 – Dec 2021

Best Data Visualization, ASA DataFest 2021 · Coursework: machine learning, distributed computing, multithreading, database systems.